

SP500 Tracker

API Documentation

Base URL (Production): <https://sp500-tracker.onrender.com>

Base URL (Local): <http://localhost:8000>

Interactive Swagger UI: <https://sp500-tracker.onrender.com/docs>

OpenAPI JSON: <https://sp500-tracker.onrender.com/openapi.json>

1. Overview

The SP500 Tracker API is a RESTful web service built with FastAPI that provides real-time and historical S&P 500 stock data, user authentication via JWT, watchlist management with computed analytics, portfolio management with holdings tracking, and news aggregation via Google News RSS. All responses are JSON. The API uses standard HTTP methods (GET, POST, PATCH, DELETE) and returns appropriate status codes.

2. Authentication

The API uses Bearer Token (JWT) authentication. Workflow: 1) Register via POST /auth/register, 2) Login via POST /auth/login to receive an access_token, 3) Include the header 'Authorization: Bearer <access_token>' on subsequent requests. Tokens expire after 60 minutes. All /watchlists/* and /portfolios/* endpoints require authentication. All /stocks/* endpoints are public.

3. Error Handling

The API returns standard HTTP status codes with JSON error bodies:

Status	Meaning	Example
200	Success	Successful GET, PATCH, DELETE
201	Created	Successful POST
401	Unauthorized	Invalid or missing token
403	Forbidden	Account inactive
404	Not Found	Resource does not exist
409	Conflict	Duplicate resource
422	Unprocessable	Validation error

4. Endpoints - Auth

4.1 POST /auth/register

Create a new user account. Request body: email (string, required), password (string, 8-128 chars, required).

```
curl -X POST https://sp500-tracker.onrender.com/auth/register \
  -H "Content-Type: application/json" \
  -d '{"email": "user@example.com", "password": "securepass123"}'
```

Response 201: {"message": "Registration successful...", "user_id": 42}

4.2 POST /auth/login

Authenticate and receive a JWT token. Request body: email, password.

```
curl -X POST https://sp500-tracker.onrender.com/auth/login \
  -H "Content-Type: application/json" \
  -d '{"email": "user@example.com", "password": "securepass123"}'
```

Response 200: {"access_token": "eyJhbG...", "token_type": "bearer"}

4.3 GET /auth/me

Return the authenticated user's profile. Requires Bearer token.

```
curl https://sp500-tracker.onrender.com/auth/me \
-H "Authorization: Bearer <token>"
```

```
Response 200: {"id": 42, "email": "user@example.com", "is_active": true, ...}
```

5. Endpoints - Stocks

5.1 GET /stocks

Discover/search tickers (paginated). Query params: search (string), limit (int, default 20), offset (int, default 0). Supports fuzzy matching on ticker and company name.

```
curl "https://sp500-tracker.onrender.com/stocks?search=apple&limit=5"
```

```
Response 200: [{"ticker": "AAPL", "company_name": "Apple Inc.", ...}]
```

5.2 GET /stocks/{ticker}

Summary card with latest price, 52-week high/low, percentage changes (1W, 1M, 1Y).

```
curl https://sp500-tracker.onrender.com/stocks/AAPL
```

```
Response 200: {"ticker": "AAPL", "company_name": "Apple Inc.",
  "latest_close": 178.50, "week_52_high": 199.62, ...}
```

5.3 GET /stocks/{ticker}/history

OHLCV history. Query params: timeframe (1W|1M|3M|6M|1Y|5Y|MAX), or start_date + end_date (YYYY-MM-DD).

```
curl "https://sp500-tracker.onrender.com/stocks/AAPL/history?timeframe=1M"
```

```
Response 200: [{"date": "2025-03-01", "open": 175.0, "high": 178.0,
  "low": 174.5, "close": 177.8, "volume": 45000000}]
```

5.4 GET /stocks/{ticker}/news

News feed filtered by timeframe (1d, 7d, 30d). Aggregated from Google News RSS.

```
curl "https://sp500-tracker.onrender.com/stocks/AAPL/news?timeframe=7d"
```

```
Response 200: [{"title": "Apple announces...", "link": "https://...",
  "published": "2025-03-10T12:00:00", "source": "Reuters"}]
```

5.5 GET /stocks/{ticker}/live

Live/intraday chart data. Query params: range (1d|5d|1mo), interval (1m|5m|15m|1h|1d). Uses Finnhub (primary), Yahoo Finance (fallback), then database history (guaranteed).

```
curl "https://sp500-tracker.onrender.com/stocks/AAPL/live?range=1d&interval=5m"
```

```
Response 200: {"candles": [{"timestamp": 1710..., "open": 175.0,
  "high": 176.0, "low": 174.5, "close": 175.8, "volume": 12000}], ...}
```

6. Endpoints - Watchlists (all require auth)

6.1 POST /watchlists

Create a watchlist. Body: {"name": "Tech Stocks"}. Returns 201.

6.2 GET /watchlists

List all watchlists for the authenticated user.

6.3 DELETE /watchlists/{id}

Delete a watchlist by ID. Returns 200 with confirmation message.

6.4 GET /watchlists/{id}/items

List all tickers in a watchlist.

6.5 POST /watchlists/{id}/items

Add a ticker to a watchlist. Body: {"ticker": "AAPL"}. Returns 201. 409 if duplicate.

6.6 DELETE /watchlists/{id}/items/{ticker}

Remove a ticker from a watchlist. Returns 200.

6.7 GET /watchlists/{id}/insights

Watchlist analytics. Returns per-ticker metrics (1W/1M/1Y price change %, 30-day volatility, average volume) plus summary (top gainer, top loser, most/least volatile, concentration).

```
curl https://sp500-tracker.onrender.com/watchlists/1/insights \
-H "Authorization: Bearer <token>"
```

```
Response 200: {"items": [{"ticker": "AAPL", "change_1w_pct": 2.3,
  "volatility_30d": 0.015, ...}], "summary": {"top_gainer": "AAPL", ...}}
```

7. Endpoints - Portfolios (all require auth)

7.1 POST /portfolios

Create a portfolio. Body: {"name": "Growth"}. Returns 201.

7.2 GET /portfolios

List all portfolios for the authenticated user.

7.3 PATCH /portfolios/{id}

Rename a portfolio. Body: {"name": "New Name"}. Returns 200.

7.4 DELETE /portfolios/{id}

Delete a portfolio. Returns 200.

7.5 GET /portfolios/{id}/holdings

List all holdings in a portfolio with ticker, quantity, and average cost.

7.6 POST /portfolios/{id}/holdings

Add a holding. Body: {"ticker": "AAPL", "quantity": 10, "avg_cost": 150.00}. Returns 201.

7.7 PATCH /portfolios/{id}/holdings/{ticker}

Update a holding's quantity and/or average cost. Returns 200.

7.8 DELETE /portfolios/{id}/holdings/{ticker}

Remove a holding from a portfolio. Returns 200.

8. Complete Endpoint Summary

#	Method	Path	Auth	Description
1	POST	/auth/register	-	Create account
2	POST	/auth/login	-	Get JWT access token
3	GET	/auth/me	Yes	Current user profile
4	GET	/stocks	-	Discover/search tickers (paginated)
5	GET	/stocks/{ticker}	-	Summary card: price, % changes, 52w high/low
6	GET	/stocks/{ticker}/history	-	OHLCV history (timeframe or date range)
7	GET	/stocks/{ticker}/news	-	News feed filtered by timeframe
8	GET	/stocks/{ticker}/live	-	Live/intraday chart data
9	POST	/watchlists	Yes	Create watchlist
10	GET	/watchlists	Yes	List user's watchlists
11	DELETE	/watchlists/{id}	Yes	Delete watchlist
12	GET	/watchlists/{id}/items	Yes	List watchlist tickers
13	POST	/watchlists/{id}/items	Yes	Add ticker to watchlist
14	DELETE	/watchlists/{id}/items/{ticker}	Yes	Remove ticker from watchlist
15	GET	/watchlists/{id}/insights	Yes	Watchlist analytics & insights
16	POST	/portfolios	Yes	Create portfolio
17	GET	/portfolios	Yes	List user's portfolios
18	PATCH	/portfolios/{id}	Yes	Rename portfolio
19	DELETE	/portfolios/{id}	Yes	Delete portfolio
20	GET	/portfolios/{id}/holdings	Yes	List portfolio holdings
21	POST	/portfolios/{id}/holdings	Yes	Add holding
22	PATCH	/portfolios/{id}/holdings/{ticker}	Yes	Update holding qty/cost
23	DELETE	/portfolios/{id}/holdings/{ticker}	Yes	Remove holding

Total: 23 endpoints across 4 resource groups (Auth, Stocks, Watchlists, Portfolios).

9. Data Source

Historical OHLCV stock data sourced from:

Shahrukh, I. (2025) Top 50 S&P 500 Companies Dataset. Available at: <https://www.kaggle.com/datasets/ibrahimshahrukh/top-50-companies-dataset> (Accessed: 9 March 2025).

Live and intraday data provided by Finnhub (<https://finnhub.io/>) and Yahoo Finance

(<https://finance.yahoo.com/>).

News articles aggregated via Google News RSS (<https://news.google.com/>).